

Table 1: End-to-End Local Voice AI Receptionist Tech Stack

| Pipeline Layer   | Component / Model          | Runtime / Engine                     | Input / Output Format                                                                                                   | Primary Role                                                                                                   |
|------------------|----------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Audio I/O</b> | sounddevice (PortAudio)    | Python 3.11 C-binding                | <b>In/Out:</b> 16 kHz Mono (Mic) / 24 kHz Mono (Speaker)                                                                | Captures continuous microphone stream and plays back synthesized response audio without blocking.              |
| <b>VAD</b>       | Silero VAD (v5) + RMS Gate | onnxruntime (CPU)                    | <b>In:</b> 16 kHz Mono PCM (512-sample frames)<br><b>Out:</b> Speech probability (0.0 – 1.0) + Recurrent state          | Detects start of speech, handles silence endpointing, and triggers live <b>barge-in</b> to interrupt playback. |
| <b>STT</b>       | Faster-Whisper (base.en)   | CTranslate2 (int8 quantized)         | <b>In:</b> 16 kHz Mono audio buffer<br><b>Out:</b> Normalized text string + timestamps                                  | Converts user speech into text once VAD detects the end of an utterance.                                       |
| <b>LLM</b>       | Qwen2.5 (3B)               | Ollama (llama.cpp / Apple Metal GPU) | <b>In:</b> Conversation history + JSON tool schema<br><b>Out:</b> Conversational response tokens / structured tool call | Manages clinic reasoning, understands intent, answers FAQs, and triggers deterministic booking functions.      |
| <b>TTS</b>       | Kokoro-82M (v1.0)          | onnxruntime (CPU)                    | <b>In:</b> Plain text / phonemes (espeak-ng)<br><b>Out:</b> 24 kHz Mono float32 PCM audio                               | Synthesizes conversational voice output to stream back to the caller.                                          |